

AGENTIVE SOLUTIONS · RAG FOR AUTOMATION · LESSON 1

What is RAG & Why Businesses Need It

The AI that reads your data before it answers.

Zero prerequisites

5-step architecture

Live RAG demo

SolutionsWithShahzain · Agentive Solutions

THE PROBLEM

Ask ChatGPT About Your Own Business



"What is my shop's refund policy?"



It doesn't know — so it guesses.

Your policy document was never on the internet, so it was never in the training data.

The information a business needs most is exactly what a public model has never seen.

IN THIS LESSON

What You'll Learn



Why LLMs fail on your data

Three reasons — and why confidence makes it worse.



What RAG is, and its 5 steps

Ingest, embed, store, retrieve, generate — named and understood.



Choose the right approach

RAG vs fine-tuning vs long context — for a real client.



Where RAG makes money

The use cases clients actually pay for.

The New Employee



Day one on your shop floor

Brilliant. Has read every book in the world.
Knows nothing about your shop.



"Do you deliver same-day in Karachi?"



"Absolutely — it'll reach you in 2 hours!"



You don't offer same-day delivery. You never did.

He wasn't lying. He simply never saw your delivery policy.

WHY IT HAPPENS

Three Reasons LLMs Fail on Business Data



Stale knowledge

Training stopped on a date. Everything after it is unknown.

New products missing.



Hallucination

Doesn't know the answer, so it invents one — with full confidence.

Wrong price. Wrong policy.



No access to your data

Your documents, policies and prices were never in the training set.

Can't answer — so it guesses.

RAG solves the first and third outright — and sharply reduces the second.

THE ANSWER

What is RAG?

Retrieval-Augmented Generation



Before answering, the AI first **retrieves** the relevant facts from **your** data — then **generates** the answer using only those facts.

Retrieve first. Then answer. That one change is the whole idea.

THE NAME, UNPACKED

Retrieval · Augmented · Generation



Retrieval

Find the chunks in your data that relate to the question.



Augmented

Add those chunks into the model's prompt, alongside the question.



Generation

Write the answer using only those chunks — not memory.

chunk — a piece of a document. We keep the English term throughout the series.

The Doctor and Your File



LLM ALONE

Prescribes from memory

Never opens your file. Doesn't see your reports, allergies or current medicine — so the answer is a guess.



RAG

Opens your file first

Reads your reports, then prescribes — and tells you why: *"your report says this."*

Same medical college. Same knowledge. One difference — Doctor B read your record before answering.

That file is your business data. "Your report says this" is the citation.

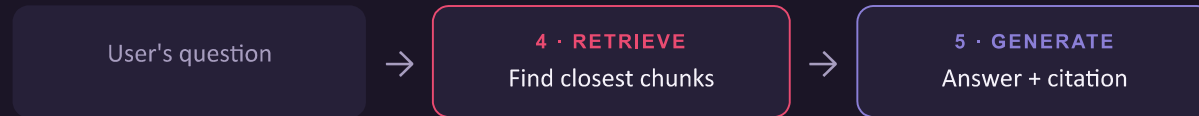
THE ARCHITECTURE

The 5 Steps of Every RAG System

PHASE 1 — INGESTION · WHEN CONTENT CHANGES



PHASE 2 — QUERY · EVERY QUESTION



Steps 1–3 run when content changes. Steps 4–5 run on every single question.

THREE WORDS YOU JUST MET

Chunk · Embedding · Vector Database



Chunk

A piece of a document.

20-page policy → ~60 chunks



Embedding

That chunk's meaning, turned into numbers.

"refund" sits near "paisa wapas"



Vector database

Stores those numbers, finds the closest ones fast.

Question in → top 3 chunks out

That's enough for today. Lesson 2 opens embeddings; Lesson 3 opens chunking.

NOT JUST THEORY

The Same 5 Steps, in Our Live Tutor

STEP	IN OUR TUTOR	DETAIL IN
1 · Ingest	Book lessons split by heading	Lesson 3
2 · Embed	OpenAI text-embedding-3-small · 1536 dims	Lesson 2
3 · Store	Qdrant Cloud, free tier	Lesson 2
4 · Retrieve	Agent tool-call + relevance filter	Lesson 4
5 · Generate	gpt-4o-mini + citations	Lesson 5

Every RAG system you will ever build is these same five steps.

LIVE

The RAG System on This Page

Everything you just learned — running in the corner of the book.

1

"What is the difference between Save and Publish in n8n?" → sources appear

2

"Hi" → no sources. It decided retrieval wasn't needed.

3

Ask in Roman Urdu → it answers in Roman Urdu.

Sources under an answer are the proof that step 4 actually ran.

Three Ways to Give AI Your Data

FINE-TUNING

Send them on a course

Their manner changes. Their knowledge doesn't.

LONG CONTEXT

Make them read the whole cabinet

Every question. Every time. You pay for all of it.

RAG

Give them the cabinet and an index

They pull the one file they need, when they need it.

The argument against fine-tuning is not cost. It teaches behaviour, not facts — and facts baked into weights can't be cited.

USE THIS IN THE MEETING

The Decision Framework

THE CLIENT'S SITUATION	YOUR ANSWER
Q&A over docs, policies, FAQs	RAG
The data changes every week	RAG
Answers must show their source	RAG
Change the tone or format, not the facts	Fine-tuning
One small document, analysed once	Long context

For business knowledge, the first answer is always RAG.

WHERE THE MONEY IS

Four Use Cases Clients Pay For



Customer support bot

Answers from policies and FAQs, 24/7. Support hours drop.



Internal knowledge assistant

HR policies, SOPs, training docs. Onboarding stops eating senior time.



Docs & education assistant

Exactly what this book runs. Students self-serve; engagement rises.



Sales & product Q&A

Catalogue and specs. A visitor who gets an answer converts.

The Travel Agency in Lahore



The problem

200+ visa and package documents. The same questions all day on WhatsApp. Two staff members answer nothing else.



The build

Documents → chunks → embeddings → vector database. The bot retrieves the right section and cites it.



The result

Staff handle bookings and complex cases only. Rates change? Re-embed one document.

The model is never retrained. That is the real business case.

AVOID THESE

Common Beginner Mistakes

THE MISTAKE

"Upload your files to ChatGPT — that's RAG"

Using RAG for counts and totals

Assuming the client's knowledge is written down

Making the whole document one chunk

Believing RAG ends hallucination

THE RIGHT APPROACH

That's one session. Production RAG is your own pipeline, under your control.

That's a database question. Vector search finds meaning — it cannot count.

Usually it isn't. Content collection is phase one — scope and price it.

Retrieval collapses. Chunking strategy decides answer quality.

It reduces it. You still need grounding rules and citations.

Three Rules That Win Projects

1

Open with the decision framework

Recommending the right approach — even when it's the cheaper one — is what separates a consultant from a tool-seller.

2

Always show citations

Trust is the product. An answer without a source is a confident guess with extra steps.

3

Sell outcomes, not architecture

Clients buy fewer support hours and faster onboarding. Nobody buys a vector database.

Five Things to Remember

- 1 LLMs fail on business data for three reasons: stale knowledge, hallucination, no access to your documents.
- 2 RAG means retrieve first, then generate — with a citation, like a doctor reading your file.
- 3 Five steps run every RAG system: ingest, embed, store, retrieve, generate.
- 4 Fine-tuning teaches behaviour, not facts — and facts in weights can't be cited.
- 5 RAG only reads what is written down. Missing content is the first blocker on real projects.

YOUR TURN

Map Three Businesses

Pick three businesses you know — or want to work with — and fill this in:

BUSINESS	WHAT DATA EXISTS?	WHO ASKS?	WHAT RAG SOLVES
Travel agency, Lahore	200+ package documents	WhatsApp customers	Repetitive inquiries handled

Post your table in the comments — the best use case goes in the next video.

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Agentic Solutions

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